SoulScope Acoustic Parameter Registry v0.1

Status: Canonical implementation companion — final for adoption
Date: August 14, 2026
Governing authority: SoulScope Canon v1.3
Scope: Voice-only production observation layer

1. Purpose

The SoulScope Acoustic Parameter Registry defines the measurements that may enter the production Evidence Engine.

Its purpose is not to maximize feature count. Its purpose is to create a measurement layer that is:

- reproducible;
- versioned;
- interpretable;
- quality-gated;
- temporally resolved;
- compatible with personal and within-session comparison;
- traceable into semantic reasoning;
- usable by the Resonance Signature without importing semantic claims.

A parameter does not become scientifically meaningful merely because it can be extracted from audio.

Every production parameter must have:

1. a permanent identifier;
2. an operational definition;
3. units;
4. extractor provenance;
5. validity conditions;
6. quality requirements;
7. supported task types;
8. known confounds;
9. reference compatibility;
10. permitted Evidence uses;
11. renderer eligibility;
12. version information.

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2. Registry architecture

SoulScope maintains four distinct measurement tiers.

Tier	Purpose	Production status
A — Time-resolved acoustic descriptors	Describe local acoustic behavior through time	Canonical
B — Prompt-level acoustic functionals	Summarize qualified acoustic trajectories	Canonical
C — SoulScope-native temporal and speech-organization parameters	Capture properties especially important to the three-prompt protocol	Canonical
Q — Quality and validity variables	Determine whether measurements may be interpreted	Canonical gating metadata
R — Research feature bank	Exploratory features, embeddings, large feature spaces	Research only

Research features may not influence a production result until promoted through registry governance and validation.

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3. Approximate parameter count

SoulScope does not use feature count as a scientific claim.

Registry v0.1 contains:

- **25 standardized time-resolved descriptor channels** based on the eGeMAPSv02 low-level descriptor specification;
- **88 standardized prompt-level functionals** based on eGeMAPSv02;
- **16 SoulScope-native acoustic/temporal parameters**;
- **8 independent signal-quality and task-validity variables**.

This produces:

$$25 + 88 + 16 = 129$$

registered acoustic parameter definitions, plus:

$$8$$

quality/validity variables.

Thus the complete production measurement record presently exposes approximately:

$$\boxed{137 \text{ registered measurement fields}}$$

before prompt contrasts, personal deviations, trajectories, or Evidence Marker derivations are counted.

This number is descriptive, not promotional.

A change in registry size does not by itself constitute scientific advancement.

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4. Tier A — Time-resolved acoustic descriptor channels

The canonical standardized core follows the eGeMAPSv02 descriptor families.

4.1 Energy and spectral structure

- `AC_LLD_LOUDNESS`
- `AC_LLD_ALPHA_RATIO`
- `AC_LLD_HAMMARBERG`
- `AC_LLD_SLOPE_0_500`
- `AC_LLD_SLOPE_500_1500`
- `AC_LLD_SPECTRAL_FLUX`

4.2 Cepstral descriptors

- `AC_LLD_MFCC1`
- `AC_LLD_MFCC2`
- `AC_LLD_MFCC3`
- `AC_LLD_MFCC4`

MFCCs are eligible for statistical/model use but are not sufficient by themselves to generate a user-facing explanation.

4.3 Fundamental frequency and voice quality

- `AC_LLD_F0_ST`
- `AC_LLD_JITTER_LOCAL`
- `AC_LLD_SHIMMER_LOCAL_DB`
- `AC_LLD_HNR`
- `AC_LLD_H1_H2_REL`
- `AC_LLD_H1_A3_REL`

4.4 Formant structure

- `AC_LLD_F1_FREQ`
- `AC_LLD_F1_BW`

- `AC_LLD_F1_AMP_REL_F0`
- `AC_LLD_F2_FREQ`
- `AC_LLD_F2_BW`
- `AC_LLD_F2_AMP_REL_F0`
- `AC_LLD_F3_FREQ`
- `AC_LLD_F3_BW`
- `AC_LLD_F3_AMP_REL_F0`

These channels are stored as qualified time series:

$$x_f(t)$$

with timestamp, validity, and extractor provenance.

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5. Tier B — Standardized prompt-level functionals

SoulScope adopts the **88-function eGeMAPSv02 functional specification by version**, rather than manually renaming or silently altering its definitions.

The implementation manifest SHALL preserve:

```
feature_specification = eGeMAPSv02
feature_level = Functionals
expected_field_count = 88
extractor_implementation
extractor_version
configuration_hash
```

These functionals cover standardized summaries of:

- F0;
- loudness;
- spectral flux;
- MFCC 1–4;
- jitter;
- shimmer;
- HNR;
- harmonic relationships;
- formant frequency;
- formant bandwidth;
- formant amplitude;
- spectral balance;
- voiced/unvoiced temporal organization;
- equivalent sound level.

SoulScope SHALL maintain an internal snapshot of the exact 88-field manifest corresponding to the adopted specification.

No extractor upgrade may silently change the historical interpretation of a stored scan.

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6. Tier C — SoulScope-native acoustic parameters

The following parameters supplement the standardized core because they are especially important to SoulScope's three 30-second spontaneous-speech conditions.

6.1 Cepstral peak prominence

`SS_CPP_MEAN`

Mean qualified cepstral peak prominence over compatible voiced segments.

`SS_CPP_VARIABILITY`

Robust within-prompt variability of qualified CPP.

Recommended statistic:

$$MAD(CPP)$$

or a calibrated robust equivalent.

`SS_CPP_DRIFT`

Robust temporal slope of qualified CPP across the prompt.

6.2 Response initiation

`SS_RESPONSE_ONSET_LATENCY`

Time from prompt-ready/capture-start event to first qualified speech onset.

Unit:

$$ms$$

Interpretation is task-bound. It may contribute to timing evidence but may not independently indicate hesitation, avoidance, deception, anxiety, or cognitive impairment.

6.3 Pause organization

`SS_PAUSE_LOAD`

$$\text{Pause Load} = \frac{\text{qualified silent-pause duration}}{\text{qualified response interval}}$$

`SS_PAUSE_MEDIAN`

Median duration of qualified silent pauses.

`SS_PAUSE_P90`

90th percentile qualified pause duration.

`SS_PAUSE_RATE`

Number of qualified pauses per unit of qualified speaking interval.

Thresholds defining a pause SHALL be versioned and validated.

6.4 Speech-run organization

`SS_RUN_MEAN`

Mean duration of continuous qualified speech runs.

`SS_RUN_VARIABILITY`

Robust variability of speech-run duration.

6.5 Rate

`SS_SPEECH_RATE`

Estimated spoken units per unit total response time, using the approved speech-rate estimator.

`SS_ARTICULATION_RATE`

Estimated spoken units per unit phonated/articulated speech time excluding qualified silent pauses.

Speech-rate and articulation-rate implementations SHALL identify their unit basis and language dependencies.

6.6 Within-prompt drift

`SS_F0_DRIFT`

Robust temporal trend in qualified F0.

`SS_LOUDNESS_DRIFT`

Robust temporal trend in qualified loudness.

`SS_SPECTRAL_FLUX_DRIFT`

Robust temporal trend in spectral flux.

`SS_RATE_DRIFT`

Robust temporal trend in speaking/articulation rate calculated over protocol-approved local windows.

Drift describes movement during the observed response.

It does not independently represent deterioration, improvement, recovery, escalation, or emotional change.

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7. Tier Q — Signal-quality and validity variables

Quality metadata determines whether a measurement is eligible to enter Evidence reasoning.

`Q_VALID_VOICED_SECONDS`

Total qualified voiced material available for compatible measurements.

`Q_VOICED_RATIO`

Proportion of the relevant response interval classified as qualified voice.

`Q_SNR`

Versioned estimate of signal-to-noise quality.

`Q_CLIPPING_RATIO`

Proportion of signal affected by clipping or saturation.

`Q_VAD_CONFIDENCE`

Aggregate confidence in speech/non-speech segmentation.

`Q_F0_CONFIDENCE`

Aggregate confidence in pitch tracking where applicable.

`Q_DEVICE_COMPATIBILITY`

Compatibility status of acquisition conditions with the active extraction protocol.

`Q_TASK_COMPLIANCE`

Whether the recording provides sufficient material consistent with the administered prompt.

Task compliance is not an evaluation of whether the person's answer was appropriate, truthful, emotionally congruent, or socially desirable.

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8. Time representation

SoulScope SHALL retain the qualified temporal sequence whenever scientifically and operationally appropriate.

The canonical measurement representation is:

$$M_f(t) = \{t_0, t_1, x_f, q_f, v_f\}$$

where:

- t_0, t_1 are temporal bounds;
- x_f is the measured value;
- q_f is measurement quality;
- v_f is validity state.

The time-resolved measurement record supports two separate downstream uses:

$$M_f(t) \rightarrow \text{Resonance Signature}$$

and

$$M_f(t) \rightarrow \text{qualified summaries / Evidence Engine.}$$

The renderer and semantic engine therefore share measurement provenance but not interpretation logic.

9. Three-prompt measurement contract

The production protocol contains three spontaneous-speech conditions of approximately 30 seconds each.

`P1_OPEN_REFERENCE`

Speak about yourself, your day, or whatever comes to mind.

Scientific role:

open spontaneous within-session reference condition

It SHALL NOT be labeled a neutral emotional baseline.

`P2_TROUBLING_CONTEXT`

Speak about something troubling you.

Scientific role:

personally salient troubling-material condition

It SHALL NOT be automatically labeled negative emotion, stress, anxiety, distress, trauma, or threat.

`P3_FUTURE_CONTEXT`

Speak about hopes or goals for the future.

Scientific role:

personally salient future-oriented condition

It SHALL NOT be automatically labeled positive emotion, optimism, motivation, confidence, or recovery.

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10. Prompt-level representation

For compatible parameter f:

$$x_{f,1}, \quad x_{f,2}, \quad x_{f,3}$$

are preserved independently.

The Contrast Engine may calculate:

$$\Delta_{21,f} = x_{f,2} - x_{f,1}$$

$$\Delta_{31,f} = x_{f,3} - x_{f,1}$$

$$\Delta_{32,f} = x_{f,3} - x_{f,2}.$$

These are **derived comparisons**, not additional biological measurements.

Each contrast SHALL preserve:

```
source_prompt_a
source_prompt_b
source_measurement_ids[]
reference_scope
quality
compatibility
uncertainty
```

---

11. Reference scopes

Every standardized deviation must state which reference was used.

`WITHIN_PROMPT`

Local behavior across a single response.

`WITHIN_SESSION`

Comparison among P1, P2, and P3.

`PERSONAL_PROVISIONAL`

Early personal reference with insufficient history for established-baseline status.

`PERSONAL_ESTABLISHED`

Version-compatible longitudinal personal reference meeting the active baseline protocol.

`POPULATION_REFERENCE`

Validated compatible reference sample.

`ABSENT`

No legitimate reference exists.

Missing personal reference SHALL NOT be replaced silently by a population reference.

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12. Robust personal deviation

Where compatible with the registered feature and reference:

$$z_f = \mathrm{clamp}\left(\frac{x_f - \mathrm{median}_{ref}}{\max(1.4826\,MAD_{ref}, \epsilon_f)}, -3, 3\right).$$

The result SHALL carry:

- reference type;
- baseline version;
- sample count;
- compatibility status;
- baseline trust.

No deviation is valid when the reference is incompatible, version-mismatched, or insufficient under the active rule.

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13. Feature provenance schema

Every stored canonical measurement SHALL contain at least:

```
measurement_id
scan_id
participant_id_or_pseudonymous_subject_id

prompt_id
window_start_ms
window_end_ms

feature_id
feature_registry_version
```

```
value
unit

extractor_id
extractor_version
extractor_configuration_hash

quality_score
validity_status
validity_reason

reference_compatibility

confound_flags[]

created_at
```

Raw measurement records are immutable.

Corrections produce a new version; they do not overwrite historical values.

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14. Evidence eligibility

A measurement may enter semantic reasoning only if:

1. its extractor/version is authorized;
2. the measurement is valid for the source material;
3. signal quality meets feature-specific requirements;
4. task requirements are met;
5. required reference conditions are satisfied;
6. relevant confounds are recorded;
7. the Evidence rule consuming it is active and version-compatible.

A measured number can therefore exist while being **ineligible for semantic use**.

That distinction is intentional.

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15. Evidence Marker boundary

Raw acoustic measurements do not directly represent emotional or psychological meaning.

The following form is prohibited:

$$F0 \uparrow \Rightarrow \text{stress}$$

or:

$$CPP \downarrow \Rightarrow \text{cognitive demand.}$$

Instead:

$$\text{Measurement} \rightarrow \text{descriptive Evidence Marker} \rightarrow \text{functional parameter.}$$

Examples of eligible descriptive markers include:

```
EV_PROSODIC_EXPANSION
EV_PROSODIC_COMPRESSION
EV_ENERGY_SHIFT
EV_PAUSE_LOAD_INCREASE
EV_PAUSE_LOAD_DECREASE
EV_TEMPORAL_FRAGMENTATION
EV_OUTPUT_CONTINUITY
EV_RATE_ACCELERATION
EV_RATE_DECELERATION
EV_PHONATORY_SHIFT
EV_PHONATORY_STABILITY
EV_SPECTRAL_SHIFT
EV_MODULATION_BREADTH
EV_WITHIN_PROMPT_DRIFT
EV_CONTEXT_RECONFIGURATION
EV_CONTEXT_CONSISTENCY
```

Evidence Marker rules are maintained in a separate versioned Evidence Registry.

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16. Explainability classes

Every parameter is assigned one of three explainability classes.

`E1 — Directly explainable`

Suitable for evidence-level user disclosure.

Examples:

- pause duration;
- speaking rate;
- pitch range;
- voiced duration.

`E2 — Explainable with translation`

Technically interpretable but should usually be translated.

Examples:

- CPP;
- HNR;
- spectral tilt.

`E3 — Model/research only`

Useful statistically but insufficiently intuitive or physiologically specific for direct user explanation.

Examples may include:

- MFCCs;
- learned embeddings;
- high-dimensional latent features.

An `E3` variable may support a calibrated model only where policy permits, but it SHALL NOT be presented as the human-readable reason for a result by itself.

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17. Renderer eligibility

The Resonance Signature may consume qualified time-resolved acoustic descriptors explicitly authorized by the current Resonance Signature Guide.

Renderer eligibility is stored separately from semantic eligibility.

A feature may therefore be:

```
renderer_use = true
semantic_use = false
```

or:

```
renderer_use = false
semantic_use = true
```

The renderer SHALL NOT turn an acoustic parameter into a psychological meaning.

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18. Research feature bank

SoulScope MAY compute substantially larger exploratory feature spaces during research.

Examples include:

- expanded openSMILE/ComParE-type feature spaces;
- alternative pitch trackers;
- glottal-source estimates;
- voice-source features;
- modulation-spectrum features;
- wavelet descriptors;
- learned acoustic embeddings;
- self-supervised speech representations;
- cross-feature interaction candidates;
- nonlinear temporal dynamics.

Research features SHALL be namespace-isolated:

```
R_*
```

and may not enter production Evidence without:

1. documented scientific rationale;
2. reliability evaluation;
3. confound analysis;
4. fairness evaluation where relevant;
5. versioned implementation;
6. approval through registry governance.

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19. Extractor independence

The canonical specification defines **what SoulScope measures**, not which vendor SoulScope must permanently use.

The implementation may use:

- a properly licensed third-party extractor;
- an independently implemented extractor;
- multiple validated extractors for cross-checking.

Any replacement extractor must pass agreement testing against the frozen canonical measurement specification.

Historical results retain their original extractor provenance.

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20. Validation states

Every production parameter has one of:

```
EXPERIMENTAL
MEASUREMENT_VERIFIED
RELIABILITY_VERIFIED
CALIBRATED
PRODUCTION_ACTIVE
DEPRECATED
```

A feature can be technically measurable without being authorized to support semantic conclusions.

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21. Registry governance

Each registry change SHALL declare:

```
change_id
feature_id
old_version
new_version
change_type
scientific_rationale
implementation_change
validation_impact
historical_compatibility
approved_by
activation_date
```

Changes affecting interpretation, thresholds, extraction mathematics, reference behavior, or validity requirements require semantic versioning.

Historical results SHALL never be silently recalculated.

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22. Hard rules

AR-01 — Measurement is not meaning.
An acoustic parameter may not directly assert an emotional, diagnostic, personality, spiritual, deception, identity, causal, or hidden-state conclusion.

AR-02 — Provenance is mandatory.
No production measurement exists without extractor and version provenance.

AR-03 — Quality precedes inference.
Invalid measurements cannot become semantic evidence.

AR-04 — Unknown remains unknown.
Missing or invalid measurement cannot be replaced by zero, average, balanced, or aesthetically convenient values.

AR-05 — Reference identity travels with every deviation.
Population, personal, and within-session references are never interchangeable.

AR-06 — Research cannot leak into production.
Experimental features cannot silently affect a user result.

AR-07 — Time is preserved.
Where the source parameter is time-resolved, prompt-level summaries do not destroy the authoritative temporal measurement record.

AR-08 — Renderer and semantic eligibility are independent.

AR-09 — Numbers are versioned.
Thresholds, windows, extractors, and normalization rules are engineering hypotheses until validated and calibrated.

AR-10 — Feature count is not a product claim.
Scientific value is determined by reliability, calibration, relevance, and traceability—not by the largest number of extracted variables.

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23. Production output contract

The Acoustic Engine produces:

```
AcousticMeasurementRecord
├── acquisition_manifest
├── extractor_manifest
```

```
├── time_resolved_measurements[]
├── prompt_functionals[]
├── soulscope_native_parameters[]
├── quality_variables[]
├── prompt_contrasts[]
├── reference_deviations[]
└── provenance_hash
```

This immutable record is the common measurement foundation for:

1. the Evidence Engine; and
2. the Resonance Signature renderer.

Neither downstream system may modify the source measurement record.

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24. North Star

> **SoulScope measures broadly, interprets conservatively, and preserves the path between the two.**

The purpose of a rich acoustic backend is not to produce more claims.

It is to provide enough independent, high-quality information that the claims SoulScope does make can be bounded, reproducible, and explained.